

Tien Dung Vu

AI ENGINEER

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Epping, NSW

PROFESSIONAL SUMMARY

AI Engineer who ships production systems end to end: LLM agent pipelines, RAG assistants, and ML models deployed to AWS behind FastAPI endpoints. Builds the data layer underneath them too, ETL/ELT into PostgreSQL, A/B and hypothesis testing, and BI reporting for non-technical stakeholders. Takes systems from stakeholder requirements through to a deployed, measured product.

WORK EXPERIENCE

AI-Driven Marketing Intern

May 2026 - Present

Affinity Marketing Australia - Sydney, Australia

- Build and maintain 4 end-to-end content automation pipelines with Claude Code and n8n (SEO blog expander, video-to-content-pack, blog-to-social repurposing, ad copy refresh), plus an email subject-line **A/B testing** pipeline and a weekly analytics summary emailer
- Built a HubSpot CRM data-entry automation system, writing the integration code to automate input and formatting of customer and contract records across 500+ contacts, cutting data-formatting error rates by 50%

AI Engineer Intern

Oct 2024 - Apr 2025

CMC Corporation - Hanoi, Vietnam

- Engineered the AI layer of an employee onboarding system: a **retrieval-augmented generation (RAG)** assistant over 700 internal policy documents (~3,000 pages), serving semantic retrieval from a **FAISS vector index** into LLM APIs with prompt engineering, so 50 HR staff and new starters could query procedures in natural language
- Deployed the trained models to production on AWS** behind **10 FastAPI endpoints** and containerised with Docker, so the system consumed predictions through a stable cloud API rather than ad hoc scripts
- Built scikit-learn and XGBoost models for propensity and attrition risk scoring, K-Means cohort segmentation, and time-series forecasts of onboarding volume, alongside **PyTorch named entity recognition** and an OCR / document AI step over 1,250 scanned documents
- Designed and built ETL/ELT pipelines in Python and SQL loading **120,000+ HR and operational records** into PostgreSQL on daily incremental loads with schema validation and deduplication, scoped with HR and operations stakeholders in a 5-person team and delivered in two-week Jira sprints

AI Engineer Intern

Sep 2023 - Feb 2024

FPT - Hanoi, Vietnam

- Designed and implemented a **GitHub Actions CI/CD pipeline** from scratch: tests and linting on every pull request and merge, then Docker image build, tag, and push to a container registry, deploying each environment under its own secrets and gates
- Built supervised models (linear regression, logistic regression, decision trees), K-Means clustering pipelines, and NLP text classification with word embeddings, benchmarking architectures in Google Colab against evaluation metrics

PROJECTS

Pacific Wings Airline Strategy Simulator

Jun 2026 - Present

Python, FastAPI, LangGraph, Gemini, XGBoost, scikit-learn, PostgreSQL, Docker, Next.js, TypeScript

- Implemented a **LangGraph five-agent pipeline** where deterministic Demand and Finance agents extract figures from the simulation engine and Market, Risk, and Strategy agents call Gemini to narrate them, plus a conversational executive agent backed by 11 function-calling tools
- Trained and validated an **XGBoost demand forecasting model** on real aviation and macroeconomic data ($R^2=0.952$, $MAPE=15.3\%$ on a held-out 2024 test set), with 5-fold cross-validation and a bootstrap-ensemble confidence score surfaced across the UI

LLM Research Agent (CLI)

Jul 2025

Python, LangGraph, Gemini (generation + embeddings), NumPy, SerpAPI, Docker, Pytest, Ruff, GitHub Actions

- Built a research agent turning a natural-language question into a cited answer as strict JSON: a LangGraph StateGraph over six nodes with a conditional edge that routes back to search with model-written follow-up queries when context misses a required fact
- Built real RAG in place of snippet-answering: concurrent page fetch, HTML extraction, overlapping chunks, lexical prefilter, batched embeddings, cosine similarity and MMR; removed a class of citation hallucination by rebuilding citations from the retrieved context by index
- Hardened every network path with timeouts, bounded reads, and backoff retry, and scored the agent on a 25-question golden set (citation validity, recall, cost, p95 latency) with an 87-test offline Pytest suite in GitHub Actions CI

Australian Media Sentiment and Framing of AI

Jan - Jun 2026

Python, VADER, RoBERTa, FinBERT, BERTopic, Guardian API | University of Technology Sydney

- Built an ETL pipeline collecting 2,622 Guardian Australia articles via the Guardian Open Platform API, filtered to 1,033 spanning 2017 to 2026, then scored sentiment with VADER, RoBERTa, and FinBERT, using a FinBERT sliding window over full article bodies plus BERTopic theme extraction, delivering 6 editorial recommendations to The Sydney Morning Herald

EDUCATION

Bachelor of Artificial Intelligence, University of Technology Sydney

Graded: High Distinction

SKILLS

Programming: Python, SQL, Java

LLM & Agentic Systems: LangGraph (LangChain ecosystem), LLM Pipeline Design, Agentic Workflow Design, Prompt Engineering, RAG, Embeddings, Vector Search, FAISS, Semantic Retrieval

Machine Learning & NLP: PyTorch, scikit-learn, XGBoost, Named Entity Recognition, OCR / Document AI, Text Classification, Sentiment Analysis, Topic Modelling, BERT, TF-IDF, Naive Bayes, Regression, Decision Trees, K-Means Clustering, Time-Series Forecasting

Cloud, Deployment & Data: AWS, Docker, FastAPI, Model Serving, Containerised Deployment, CI/CD, GitHub Actions, Container Registry, ETL/ELT Pipelines, PostgreSQL, Feature Engineering, Data Quality Assurance

Statistics & Experimentation: Hypothesis Testing, A/B Testing, Confidence Intervals, Cross-Validation, Model Evaluation Metrics (AUC-ROC, Precision/Recall/F1)

BI & Tools: Power BI, Looker Studio, GA4, Google Search Console, Advanced Excel (Power Query, Pivot Tables, XLOOKUP), Git, Jira, Agile/Scrum, REST APIs, Pytest, Google Colab, n8n, HubSpot